EE1003
Introduction to Computer I

Checker

Andy, Yu-Guang Chen
Associate Professor, Department of EE
National Central University
andyygchen@ee.ncu.edu.tw
Slides Credit: TA Jui-I Peng

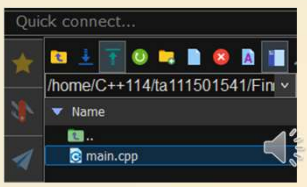
1

How to Upload Your File

- Path: /home/C++114/…..(your account)/
- Create a new folder
  - Example: ./FinalProject
- Go into the folder: $ cd your_foldername

```
[ta111501541@linuxcad30 ~]$ cd FinalProject
[ta111501541@linuxcad30 ~/FinalProject]$
```

- Upload your source code



2

How to Check Your Program

- $ cs114_final_judge </path/to source_code.cpp>
- Example:
  - $ cs114_final_judge your_filename.cpp
  - $ cs114_final_judge main.cpp
  - $ cs114_final_judge FinalProject/main.cpp
- Make sure you have the right path

```
Error: Source code not found: main.cpp
```


3

```
############################################################
Running Checks for TEST01
For Kernel size = 3x3
############################################################
Setting up input files from: /app/cases/Testcase01
Setup Success: All 7 files prepared.

================================================
Correctness Check
================================================
result_of_conv1.txt: PASS
result_of_conv2.txt: PASS
result_of_fc.txt: PASS

Relative Error after Quantization match: True
Result of the Prediction match: True

The correctness check succeeded. Starting the optimization check.

================================================
Optimization Check
================================================

--- Optimization Scoring Results ---

Energy Baseline: 192290.0
Actual Energy: 42110.0
Energy Score: 2.0

Step Baseline: 6408.0
Actual Step: 2640.0
Step Score: 2.0

--- Your optimize score of this testcase is 4.0 ---
```


4

Checker Message

- Set up

```
######################################################################
Running Checks for TEST01
For Kernel size = 3*3
######################################################################
Setting up input files from: /app/cases/Testcase01
Setup Success: All 7 files prepared.
```


Checker Message

- If your program runs longer than 6 seconds, it will be automatically terminated

```
######################################################################
Running Checks for TEST01
For Kernel size = 3x3
######################################################################
Setting up input files from: /app/cases/Testcase01
Setup Success: All 7 files prepared.
  Error: Program timed out after 6 seconds.
FATAL: Skipping validation for TEST01 due to execution failure.
FATAL: Please check your source code.
```

Checker Message

- If your program return with error code, it will show you the return code

```
########################################################################
Running Checks for TEST03
For Kernel size = 7x7
########################################################################
Setting up input files from: /app/cases/Testcase03
Setup Success: All 7 files prepared.
  Error: Program failed with return code -11.
FATAL: Skipping validation for TEST03 due to execution failure.
FATAL: Please check your source code.
```


Correctness Check

- Correctness Check Succeeded

```
########################################################################
Running Checks for TEST01
For Kernel size = 3x3
########################################################################
Setting up input files from: /app/cases/Testcase01
Setup Success: All 7 files prepared.
FAIL: return_energy() was never called

================================================
Correctness Check
================================================
result_of_conv1.txt: PASS
result_of_conv2.txt: PASS
result_of_fc.txt: PASS

Relative Error after Quantization match: True
Result of the Prediction match: True
```

- Will go to “Optimization Check” only when your correctness is correct.

Correctness Check

- Correctness Check Failed

```
================================================
Correctness Check
================================================
result_of_conv1.txt: FAIL
result_of_conv2.txt: PASS
result_of_fc.txt: PASS

Relative Error after Quantization match: True
Result of the Prediction match: True

The correctness check failed. Please review your source code.
```

- Jump to next testcase or end judging

Optimization Check

- Show the baseline and your actual result

```
================================================
Optimization Check
================================================

--- Optimization Scoring Results ---

Energy Baseline: 192290.0
Energy: 42110.0
Energy Score: 2.0

Step Baseline: 6408.0
Step: 2640.0
Step Score: 2.0

--- Your optimize score of this testcase is 4.0 ---
```

- Jump to next testcase or end judging

Optimization Check

- If got error will show the error detail

```
================================================
Optimization Check
================================================

--- Optimization Scoring Results ---

Energy Baseline: 192290.0
Actual Energy: 0.0
Energy Score: 0.0

Step Baseline: 6408.0
Actual Step: 2640.0
Step Score: 2.0

--- Your optimize score of this testcase is 2.0 ---

Error Details:
- Unable to parse energy consumption from line 3: 'Energy Consumption:'
```

- Jump to next testcase or end judging

Optimization Check

- Optimization Check Failed

```
================================================
Optimization Check
================================================
Optimization check skipped due to missing function calls.
Both return_energy() and return_step() must be called to earn optimization score.
```

- Both return_energy() and return_step() must be called to earn optimization score
- Jump to next testcase or end judging

Final Result

```
************************************************

Completed checks for all test cases.
Final Correctness Score: 10 out of 30
Final Optimization Score: 4.0 out of 12

************************************************

Kernel Size 3*3: Energy = 0.0, Step = 0
Kernel Size 5*5: Energy = 42110.0, Step = 2640
Kernel Size 7*7: Energy = 0.0, Step = 0

************************************************
```

Score

Optimization Result

Notice

- The checker works correctly for path verification, and the best solutions are verified as well. Error messages are clearly shown, so you can fix your program by yourself.

Q&A

- If you have any question about checker, please send me email
- TA: Jui I Peng
- Email: c111501541@g.ncu.edu.tw

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Thank you!